Software Requirements Specification (SRS)

AI-Powered Auto-Onboarder for ONDC (Fashion & Apparel Sellers)

Technology Stack:
Frontend: React.js, Tailwind CSS, Axios
Backend: Spring Boot 4, Spring Data MongoDB, Spring Security, Lombok
Database: MongoDB
AI Services: Gemini API, Claude API
OCR: Tess4J (Tesseract OCR)
Version Control: Git, GitHub

1. Introduction

The purpose of this project is to automate the onboarding process of Fashion and Apparel sellers onto the ONDC platform using OCR and Generative AI. The system extracts seller information from documents, generates product catalogs, validates compliance, and provides onboarding readiness reports.

2. Scope

The system supports seller registration, document upload, OCR extraction, AI-powered catalog generation, compliance validation, dashboard monitoring, and onboarding readiness assessment.

3. Existing System

Current onboarding is largely manual, requiring document verification, catalog creation, and compliance checks, resulting in delays and human errors.

4. Proposed System

The proposed solution uses Tess4J OCR to extract information from seller documents and Gemini/Claude APIs to generate product descriptions, categories, tags, and compliance suggestions automatically.

5. System Architecture

React Frontend → Spring Boot 4 Backend → MongoDB, Gemini API, Claude API, Tess4J OCR

6. Functional Requirements

FR-1 Seller Registration
FR-2 Document Upload (GST, PAN, Business License)
FR-3 OCR Data Extraction
FR-4 Product Upload
FR-5 AI Catalog Generation
FR-6 Compliance Validation
FR-7 Dashboard and Reporting

7. Non-Functional Requirements

Performance: API response under 3 seconds.
Security: JWT Authentication and BCrypt password encryption.
Scalability: Support concurrent onboarding requests.
Maintainability: Modular Spring Boot architecture.

8. Database Design

Collections: Sellers, Documents, Products, Compliance.

9. API Endpoints

/api/auth/register
/api/auth/login
/api/document/upload
/api/ocr/extract
/api/product/generate
/api/compliance/check

10. Modules

Authentication & Authorization, Seller Management, Document Management, OCR Engine Integration, AI Catalog Generator, Compliance Engine, Dashboard & Analytics.

11. Future Scope

Direct ONDC API integration, Product Image Analysis, Multi-language Support, Recommendation Engine, Bulk Upload, Cloud Deployment (AWS/Azure/GCP), Cloud Storage Integration, Advanced Analytics Dashboard.

12. Conclusion

The proposed system automates onboarding of Fashion & Apparel sellers to ONDC using Spring Boot 4, MongoDB, Tess4J OCR, Gemini API, and Claude API without requiring custom ML model development.